

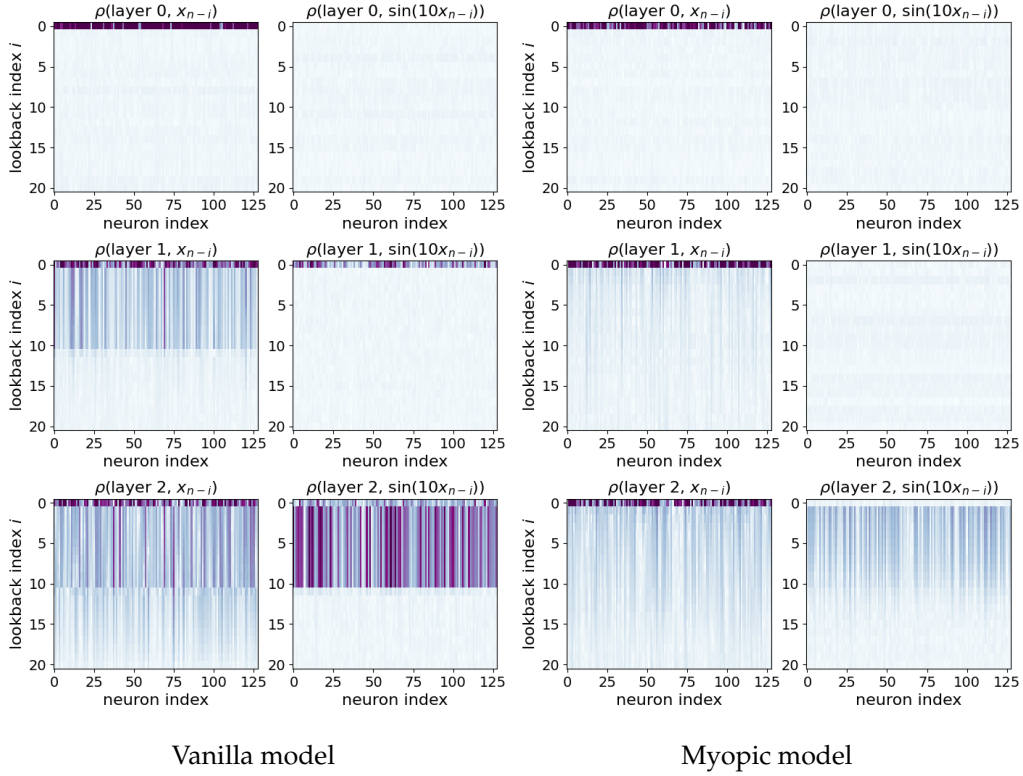


Figure 3: Empirical correlations between each hidden state neuron and x_{n-i} or $\sin(bx_{n-i})$. Models are vanilla (left two columns) and myopic (right two columns) transformers trained on $\mathcal{D}_{0.3}$.

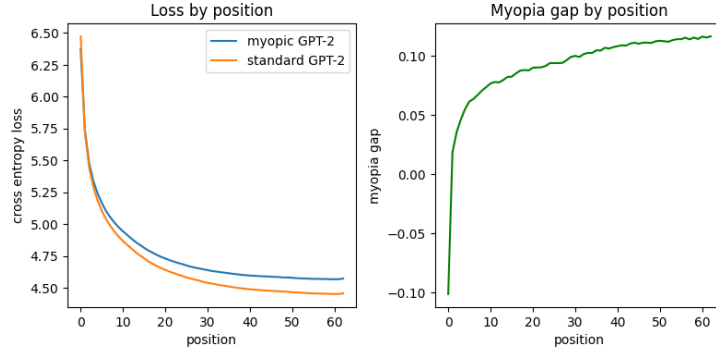


Figure 4: Cross-entropy loss of vanilla and myopic GPT-2 models by token position, and their difference. Evaluated on a sliding window over a 100K-token sample text from the PG-19 dataset (Rae et al., 2019). Aggregate cross-entropy losses on this sample are 4.67 (vanilla) and 4.77 (myopic).

5.2 Myopia gap scaling

One might suppose that the relatively small myopia gap of GPT-2 is due to the relative simplicity of the small architecture we consider, and that larger language models might exhibit a more pronounced myopia gap.

To test this, we train both vanilla and myopic transformers from the Pythia LLM suite (Biderman et al., 2023), ranging in size from 14M to 2.8B parameters, on one epoch of 10M

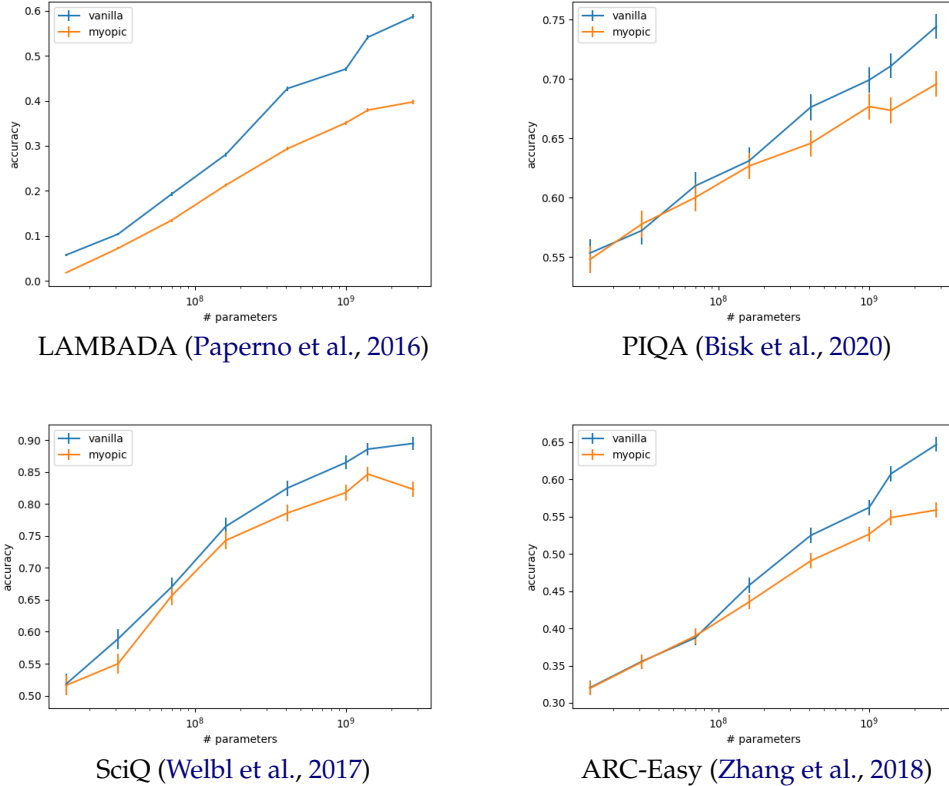


Figure 5: Benchmarks of Pythia models fine-tuned on the Pile dataset using vanilla and myopic descent.

sequences of 64 tokens each subsampled from the Pile dataset (Gao et al., 2020). (We use the same subsampled dataset for every training run.) We report validation cross-entropy loss (Figure 7 in Appendix D) as well as performance on a variety of natural language experiments (Figure 5). Note that, unlike in the GPT-2 experiments (Section 5.1), which start from random initialization, we start all training for Pythia models from the pre-trained checkpoints provided by Biderman et al. (2023)—for the larger architectures, the 10M sequence dataset we use is not sufficiently large to use for pre-training from random initialization.

6 Discussion and future work

Using a synthetic dataset, we demonstrate that pre-caching can indeed be learned by a transformer model. On the other hand, our experiments with natural language suggest that the breadcrumbs hypothesis is more explanatory for that setting, especially with smaller models, but that the importance of pre-caching increases with scale.

If the myopia gap is indeed not large in practice, there may be several applications of myopic training. We hypothesize that myopic transformers may have advantages in terms of safety and/or interpretability—it may be easier to understand what a model is doing if we know that everything that it is computing on each forward pass is directly towards the goal of predicting the immediate next token. For example, as seen in Appendix D.3, myopic models may not be able to make use of computation on the forward passes of filler tokens in the sense of Pfau et al. (2024).

Another possibility is that of automatically swapping in a locally myopic model (Section 5.1) on forward passes where we detect it is beneficial to sacrifice future performance in favor of immediate next-token accuracy (for example, on especially important tokens, or near the end of a text). We leave these possible applications to future work.

Acknowledgments

LL thanks Lukas Berglund and David Schneider-Joseph for inspiring conversations. This research was partly supported by Open Philanthropy and the Berkeley Existential Risk Initiative.

References

- Maik Barthel, Sebastian Sauppe, Stephen C Levinson, and Antje S Meyer. The timing of utterance planning in task-oriented dialogue: Evidence from a novel list-completion paradigm. *Frontiers in psychology*, 7:1858, December 2016. doi: 10.3389/fpsyg.2016.01858.
- David Bau, Jun-Yan Zhu, Hendrik Strobelt, Agata Lapedriza, Bolei Zhou, and Antonio Torralba. Understanding the role of individual units in a deep neural network. *Proceedings of the National Academy of Sciences*, 117(48):30071–30078, September 2020. ISSN 1091-6490. doi: 10.1073/pnas.1907375117. URL <http://dx.doi.org/10.1073/pnas.1907375117>.
- Amir Beck. *First-Order Methods in Optimization*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2017. doi: 10.1137/1.9781611974997. URL <https://epubs.siam.org/doi/abs/10.1137/1.9781611974997>.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances, 2021.
- Yonatan Belinkov and James Glass. Analysis methods in neural language processing: A survey, 2019.
- Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens, 2023.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. Pythia: a suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning, ICML’23*. JMLR.org, 2023.
- Yonatan Bisk, Rowan Zellers, Ronan Le bras, Jianfeng Gao, and Yejin Choi. Piqa: Reasoning about physical commonsense in natural language. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05):7432–7439, Apr. 2020. doi: 10.1609/aaai.v34i05.6239. URL <https://ojs.aaai.org/index.php/AAAI/article/view/6239>.
- Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads, 2024.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The Pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. Linearity of relation decoding in transformer language models, 2024.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks, 2019.
- Falk Huettig. Four central questions about prediction in language processing. *Brain Research*, 1626:118–135, 2015. ISSN 0006-8993. doi: <https://doi.org/10.1016/j.brainres.2015.02.014>. URL <https://www.sciencedirect.com/science/article/pii/S0006899315001146>. Predictive and Attentive Processing in Perception and Action.

- Kenneth Li, Aspen K. Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task, 2023.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt, 2023.
- George A. Miller. *Language and communication*. McGraw-Hill, New York, NY, US, 1951. doi: 10.1037/11135-000.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability, 2023.
- Yurii Nesterov. *Lectures on Convex Optimization*. Springer Publishing Company, Incorporated, 2nd edition, 2018. ISBN 3319915770.
- Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. MS MARCO: A human generated machine reading comprehension dataset. In Tarek Richard Besold, Antoine Bordes, Artur S. d’Avila Garcez, and Greg Wayne (eds.), *Proceedings of the Workshop on Cognitive Computation: Integrating neural and symbolic approaches 2016 co-located with the 30th Annual Conference on Neural Information Processing Systems (NIPS 2016), Barcelona, Spain, December 9, 2016*, volume 1773 of *CEUR Workshop Proceedings*. CEUR-WS.org, 2016. URL https://ceur-ws.org/Vol-1773/CoCoNIPS_2016_paper9.pdf.
- nostalgebraist. Interpreting gpt: The logit lens, 2020.
- Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020. doi: 10.23915/distill.00024.001. <https://distill.pub/2020/circuits/zoom-in>.
- Koyena Pal, Jiuding Sun, Andrew Yuan, Byron Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.conll-1.37. URL <http://dx.doi.org/10.18653/v1/2023.conll-1.37>.
- Denis Paperno, Germán Kruszewski, Angeliki Lazaridou, Ngoc Quan Pham, Raffaella Bernardi, Sandro Pezzelle, Marco Baroni, Gemma Boleda, and Raquel Fernández. The LAMBADA dataset: Word prediction requiring a broad discourse context. In Katrin Erk and Noah A. Smith (eds.), *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1525–1534, Berlin, Germany, August 2016. Association for Computational Linguistics. doi: 10.18653/v1/P16-1144. URL <https://aclanthology.org/P16-1144>.
- Jacob Pfau, William Merrill, and Samuel R. Bowman. Let’s think dot by dot: Hidden computation in transformer language models, 2024.
- Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. Information-theoretic probing for linguistic structure, 2020.
- Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. 2019. URL <https://api.semanticscholar.org/CorpusID:160025533>.
- Jack W Rae, Anna Potapenko, Siddhant M Jayakumar, Chloe Hillier, and Timothy P Lillicrap. Compressive transformers for long-range sequence modelling. *arXiv preprint*, 2019. URL <https://arxiv.org/abs/1911.05507>.
- Ruoqi Shen, Sébastien Bubeck, Ronen Eldan, Yin Tat Lee, Yuanzhi Li, and Yi Zhang. Positional description matters for transformers arithmetic, 2023.